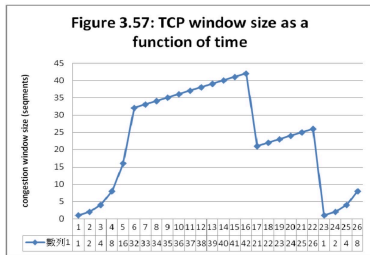


CS 2B 01357101 黃翊宏 (1人)

Computer Networking HW3

P32 Problems

Consider the following plot of TCP window size as a function of time.



Assuming TCP Reno is the protocol experiencing the behavior shown above, answer the following questions. In all cases, you should provide a short discussion justifying your answer.

- 21
- 13
- 1~6, 23~26
- cwnd = 4, ssthresh = 4
- 3個重覆ACK
- timeout
- 6~16, 17~22
- 32
- 7

P31 Problems

P31. Recall the idealized model for the steady-state dynamics of TCP. In the period of

time from when the connection's window size varies from $\frac{W}{2}$ (MSS) to W (MSS),

only one packet is lost (at the very end of the period).

(a.) show that the loss rate is equal to $L = \text{loss rate} =$

$$\frac{1}{\left(\frac{3}{8}W^2 + \frac{3}{4}W\right)}$$

(b.) Use the result above to show that if a connection has loss rate L , then its average bandwidth is approximately given by

$$\approx \frac{1.22 * MSS}{RTT \sqrt{L}}$$

a. 總 Segment 數:
$$\frac{\left(\frac{W}{2} + 1\right) \times \left(\frac{W}{2} + W\right)}{2}$$

$$= \frac{\frac{W^2}{4} + \frac{W^2}{2} + \frac{W}{2} + W}{2}$$

$$= \frac{\frac{3}{4}W^2 + \frac{3}{2}W}{2} = \frac{3}{8}W^2 + \frac{3}{4}W$$

$\therefore$ Assume 整個週期中只有一次移失

$$\therefore L = \frac{1}{\frac{3}{8}W^2 + \frac{3}{4}W}$$

b.
$$L \approx \frac{1}{\frac{3}{8}W^2} = \frac{8}{3W^2}, W \approx \sqrt{\frac{8}{3L}}$$

$$W_{AVG} = \frac{\left(\frac{W}{2} + W\right)}{2} = \frac{3}{4}W$$

$$\Rightarrow \frac{3}{4}W \times \frac{MSS}{RTT}$$

$$\frac{3}{4} \times \frac{2\sqrt{2}}{\sqrt{3L}} \times \frac{MSS}{RTT} = \frac{6\sqrt{2}}{\sqrt{48L}} \times \frac{MSS}{RTT}$$

$$= \frac{6\sqrt{2}}{4\sqrt{3}} \times \frac{MSS}{RTT \sqrt{L}} = \frac{6\sqrt{2}}{4\sqrt{3}} \approx 1.224$$